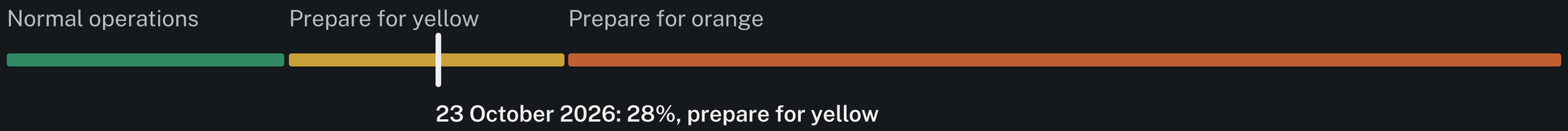
vog-desk

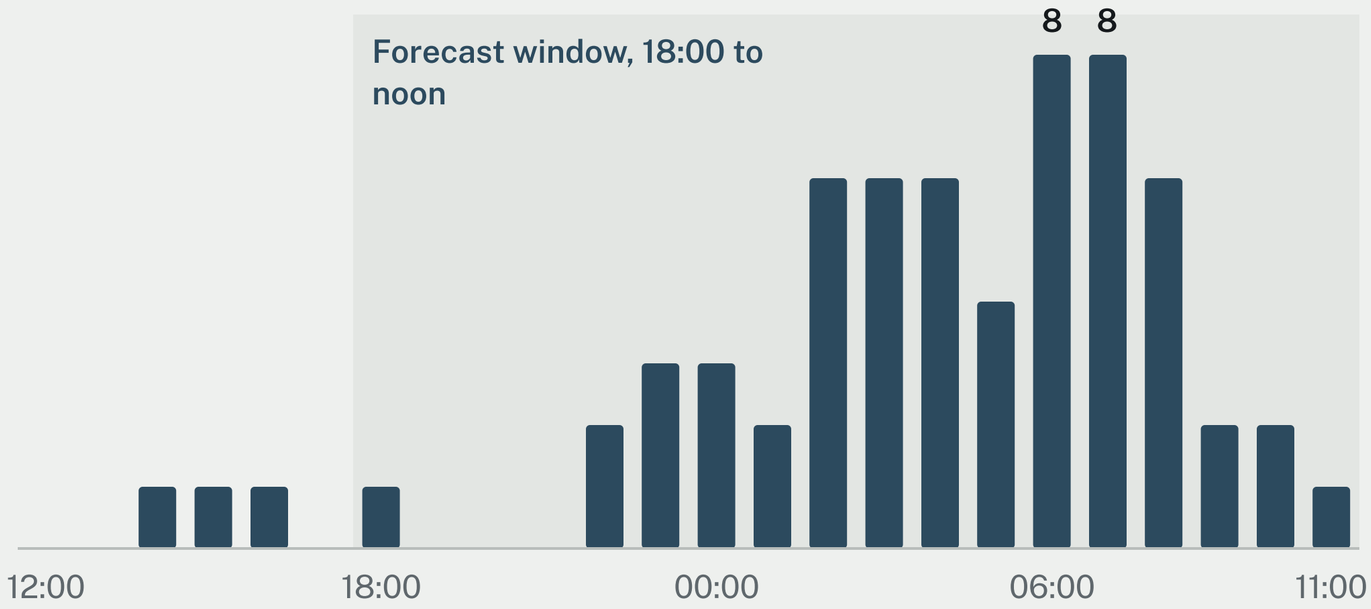
An evening-before sulfur dioxide forecast for Pāhala, Hawaiʻi Island, with the evidence behind every number.



Open Agent Hackathon 2026, Track 3. Team dotisacat.

Bad air at Pāhala comes before school

Hours at or above 75 ppb at the Pāhala monitor in 2025, by hour of day (HST)



25 days
in 2025 with an hour at or above the 75 ppb standard

60 of 63 hours
fell between 18:00 and noon, most of them between 02:00 and 09:00

After 30 minutes
of high readings the schools plan acts. It has no forecast for the evening before.

Users: the school’s designated person, the clinic, and families in Kaʻū.

Source: EPA AQS and AirNow hourly SO2 at Pāhala; Department of Health 2025 exceedance list (25 of 25 days match).

The question was fixed before any model was fit

Issued
18:00 HST
every evening, using only data public by then

Window
to 12:00 HST
the next day, eighteen hours

Event
≥ 75 ppb
in any hour at the Pāhala monitor

Score
Brier
paired differences, 7-day block bootstrap, 90% intervals

Train, 2025
356 nights, 21 events

Development, Jan to Jun 2026
tuning only

Held out, 1 Jul to 21 Oct 2026
102 nights, 39 events, scored once

Live, from 23 Oct
committed before the outcome

Pre-registration: git tag prereg, 23 Oct 2026, 15:28 PDT, before the first fit. Changes after freezing are logged in docs/PREREG.md.

What the desk does every evening

1
Evidence
Seven queries through Zetaris, cut at 18:00. Lightning summarises; a validator checks every number.

2
Four streams
The frozen 2025 model, the memory model, the recent event rate, and twenty similar remembered nights.

3
Review
Code checks the streams against each other and against the last 24 hours, and raises flags.

4
Decision
The base is the memory model. Ultra may move it by up to 0.15, only when a flag was raised.

5
Next day
Score the night against the monitor and write it to memory with source and confidence.



A flag orders targeted re-queries, at most two rounds. A disagreement that survives is stated as real; a missing GFS run sends the case to a person.

A LangGraph graph over a typed state of Pydantic records. Every step is written to a JSONL trace for the case.

The eruption changed, and the 2025 model stopped working

Measured on Pāhala nights	2025, training	Jan to Jun 2026
Share of nights with an event	5.9%	18.9%
Events during a fountaining episode	9 of 21	0 of 33
Wind toward Pāhala as a signal, AUC	0.41	0.74
Pāhala’s own last 24 hours, AUC	0.66	0.68

On the 2026 development months the 2025 model scored below climatology, a skill score of −0.03. Every Department of Health exceedance in those months is in our labels, 34 rows over 31 nights, so the shift is in the official record.

Source: docs/DEVLOG.md, development-period audit of 23 October 2026.

Memory trains the model; code sets the blend

Every verified night is remembered

H-MEM stores its features, its outcome, the data source, and a confidence that depends on how complete the hours were.

The 2025 pipeline is refit on memory each evening

Same twelve inputs, trained on the last 120 days of verified nights. With too few, all history with a 60-day half-life.

Nothing is remembered early

A night becomes visible to the desk only after it was verified, at 14:00 HST the next day.

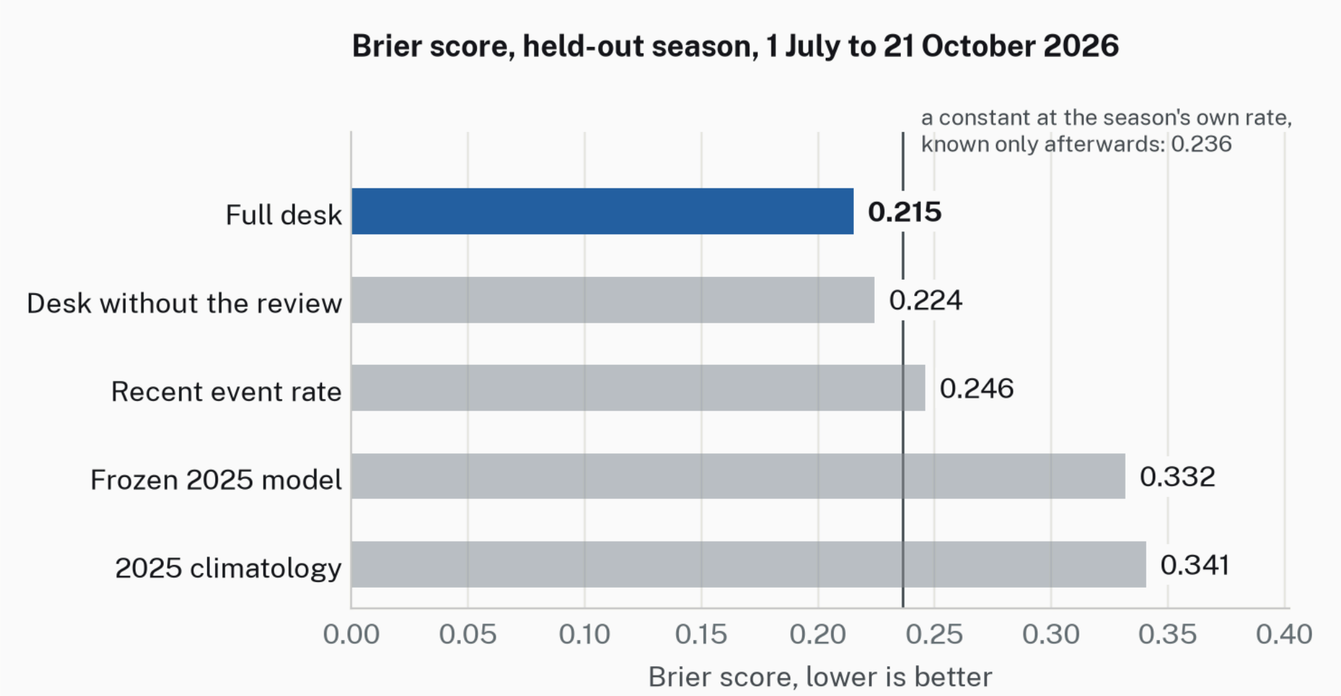
Development months, each stream alone

Stream	Brier	AUC
Memory model	0.148	0.685
Recent event rate	0.163	0.478
Similar nights, shrunk	0.175	0.431
Frozen 2025 model	0.177	0.539
Desk with Ultra’s own weights	0.162	0.575

Every mix of streams scored worse than the memory model alone, and Ultra’s weights cost 0.014. So code sets the base, and Ultra may adjust it only when a flag is raised.

Source: docs/DEVLOG.md, development backtest version 2, 173 nights.

On the held-out season, the desk beat the model it was given



Paired difference in Brier, desk minus each comparison, with 90% intervals

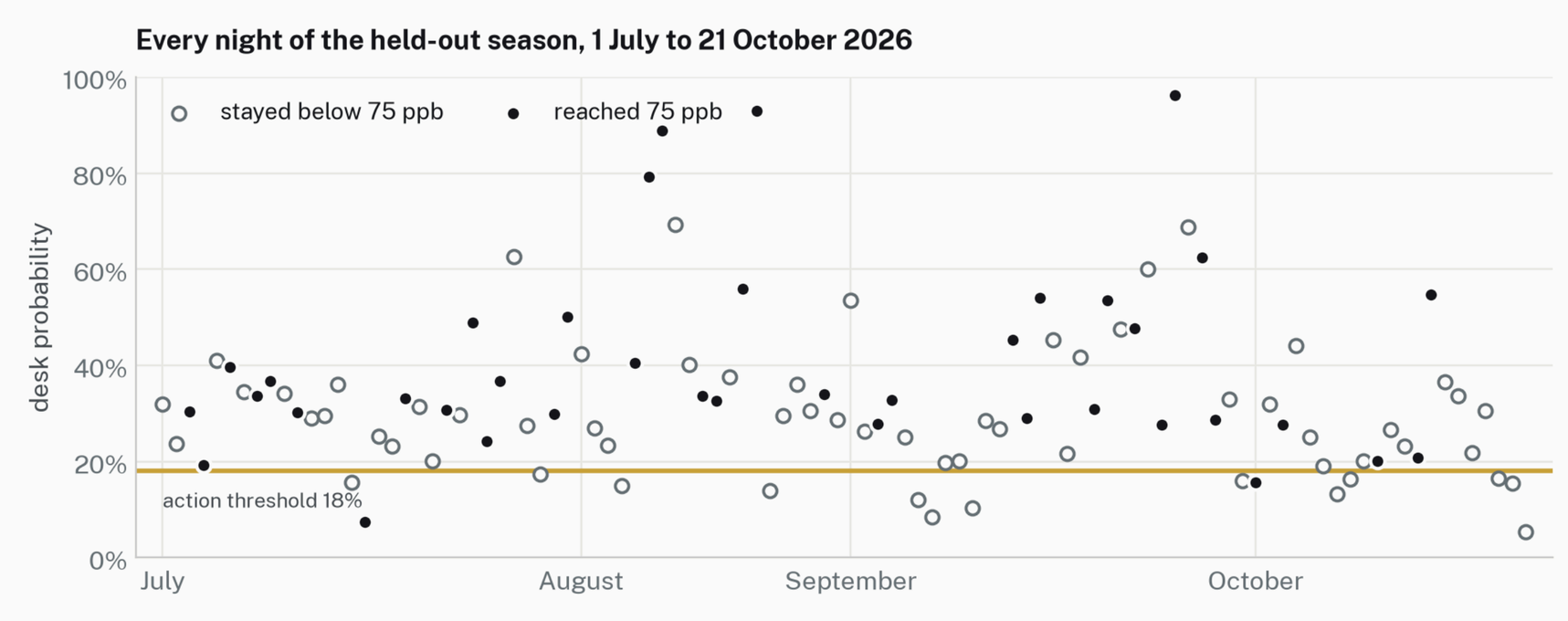
Frozen 2025 model
−0.117
[−0.157, −0.080]

Recent event rate
−0.031
[−0.048, −0.014]

Desk without the review
−0.009
[−0.019, +0.001], not established

102 nights, 39 with an event, 1 July to 21 October 2026, scored once. 7-day block bootstrap. docs/RESULTS.md

It warned on 87 of 102 nights and missed two event nights



37
hits

50
false alarms

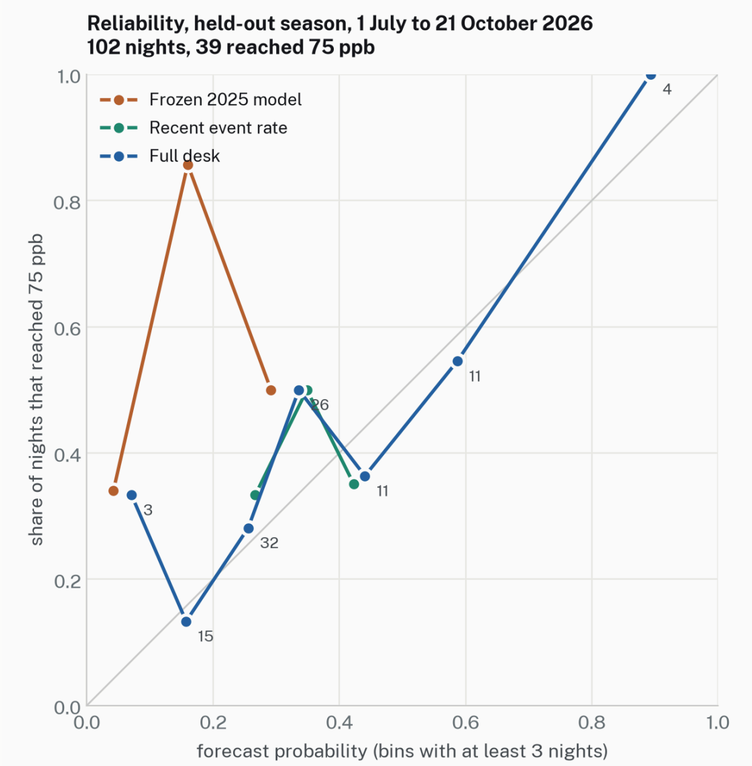
2
missed event nights

13
correct all-clears

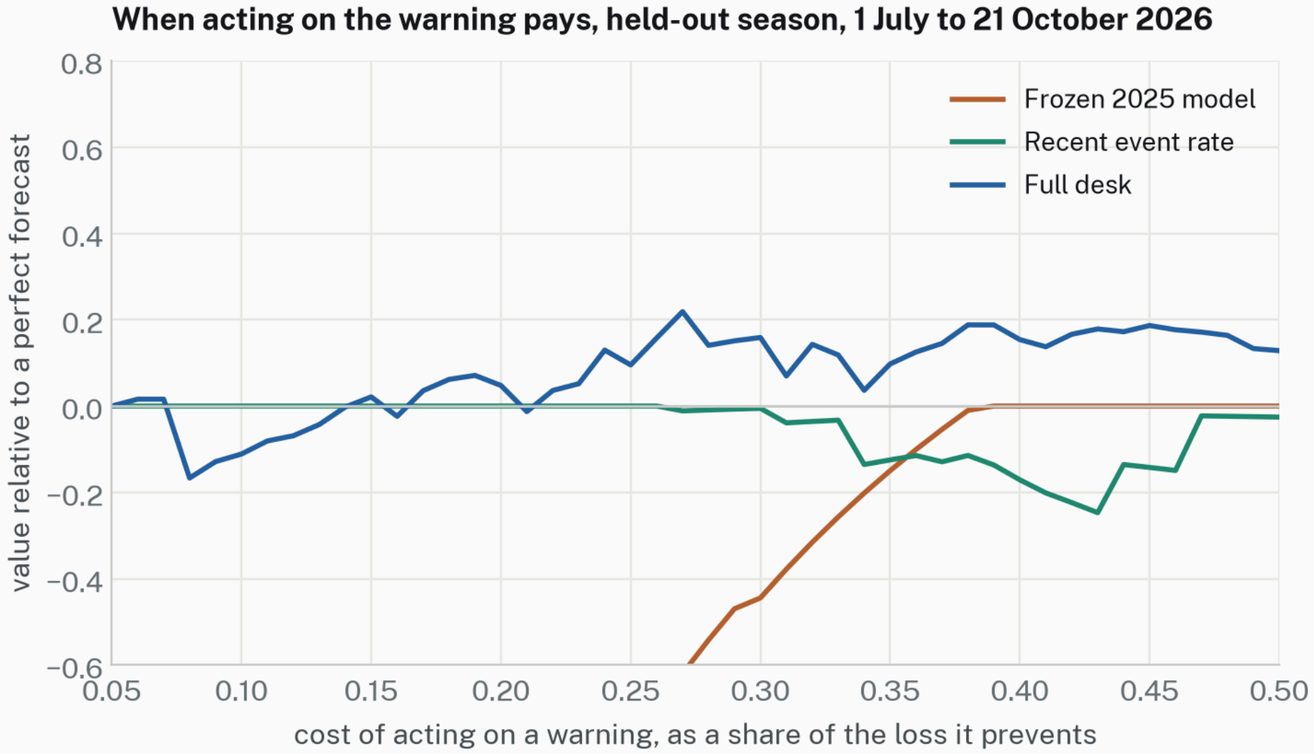
Forty of the 87 warnings were “prepare for orange”, and 19 of those 40 nights had an event.

Frozen threshold 0.18. POD 0.95, FAR 0.58, CSI 0.42, against 0.07 for the frozen model at the same threshold.

Calibrated where most nights fell



Most nights fell between 0.15 and 0.60, where the desk's bins sit within 0.16 of the diagonal. The frozen model never went above 0.30.

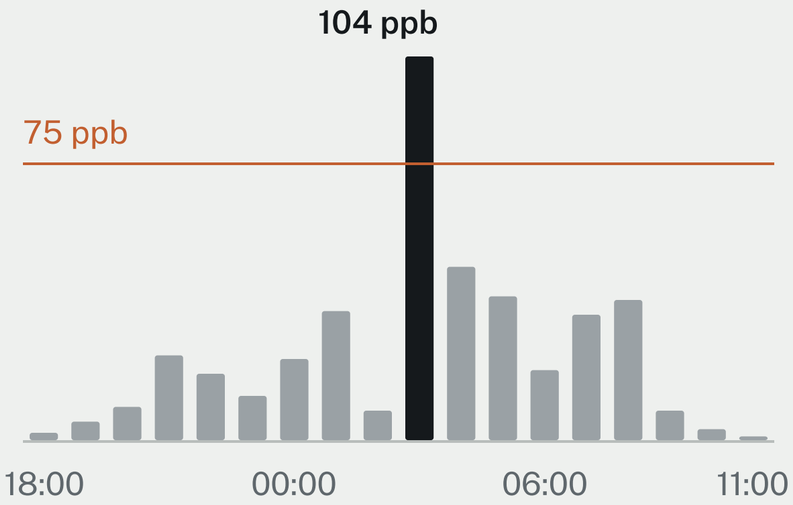


Above a cost ratio near 0.22, acting on the desk's warning beats always acting and never acting. Below 0.15 it mostly does not.

One held-out night, 31 July 2026

18:00	Seven queries through the semantic layer, each cut at the issue time. Pāhala's last 24 hours peaked at 80 ppb; episode 52 ended 2.6 days ago.
Summary	Lightning writes the regime summary in 4.9 s. Every number in it matched the evidence.
Streams	Frozen model 0.03, memory model 0.60, recent rate 0.37, similar nights 0.53 (8 of 12 reached 75 ppb).
Review	Two flags: the models disagree and the similar nights split. Four tables are re-queried and both flags close.
Decision	Ultra lowers the base by 0.10 to 0.50, citing near-zero upwind readings [E3]. Code maps 0.50 to prepare for orange.
Next day	104 ppb at 03:00 HST, a hit. The cut cost Brier: 0.248 against 0.159 for the base.

Pāhala, hourly SO2, 18:00 to 11:00 HST



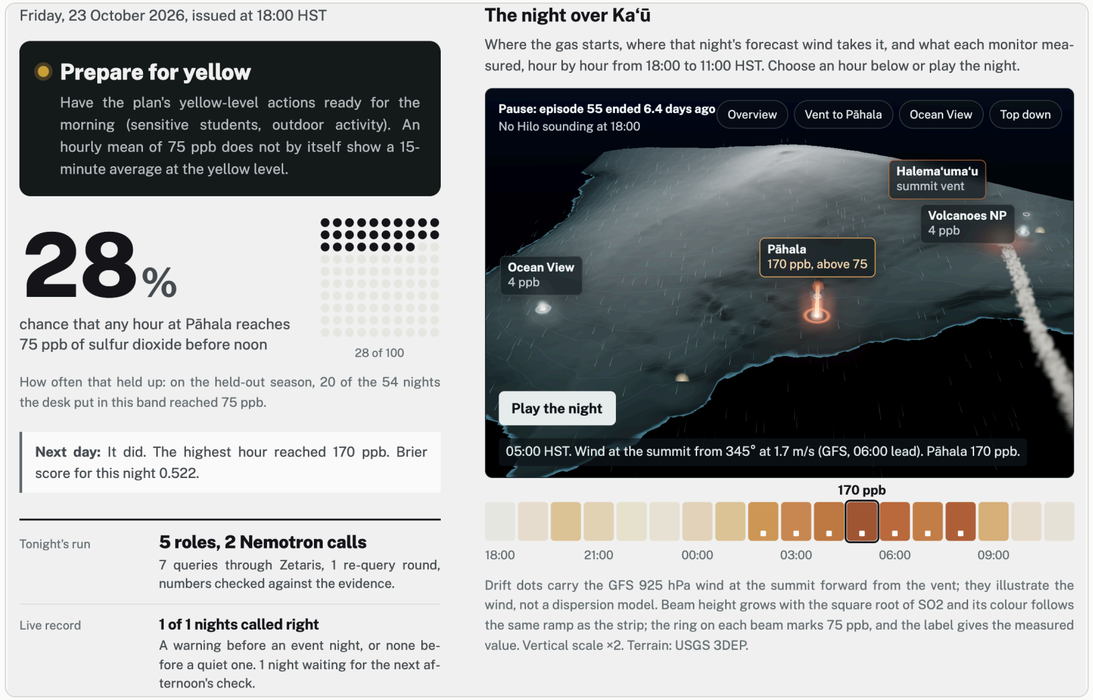
Trace: logs/backtest/full_v4, case bt-heldout-full_v4-2026-07-31. Both Ultra calls passed validation.

Sponsor technology and what it does here

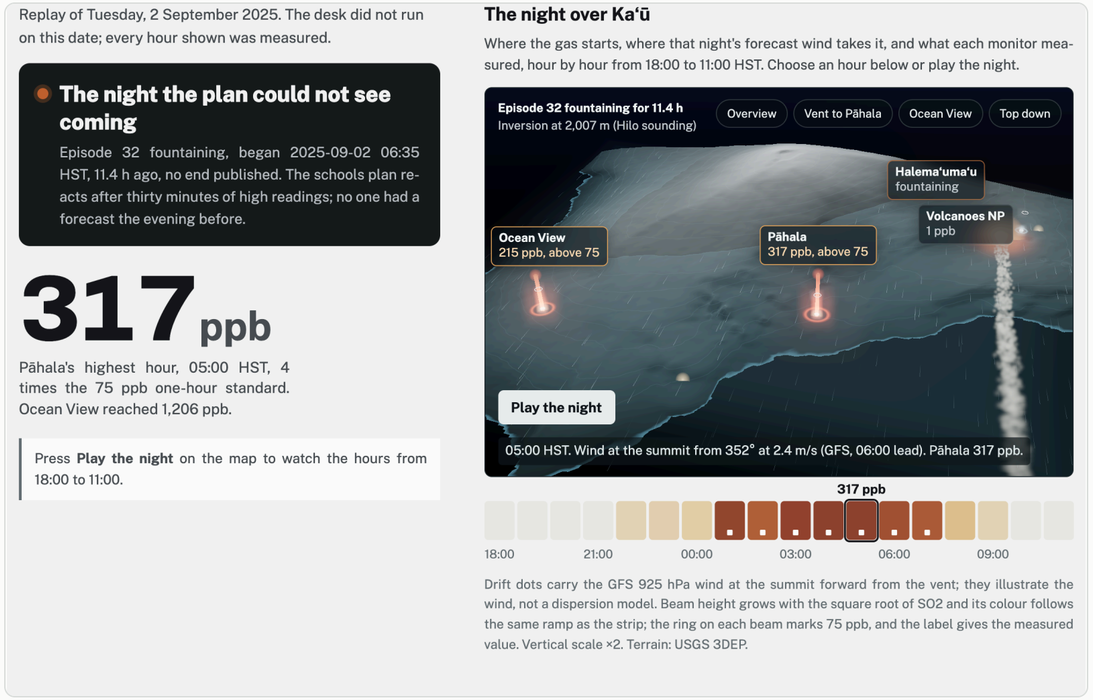
Part	Role in the desk
Zetaris	One semantic layer, pahala_exposure, over eight sources in S3 Parquet and Postgres. Its five time-stamped tables are cut at the issue time (available_utc, or when each eruption episode's start and end were published), and eleven data quality rules sit on the eight tables. Agents read only through run_sql, cut at the issue time.
Meterless	H-MEM and World Model reference engines, unchanged, behind a sidecar that adds the as-of rule. Conformance 13 of 13 and 8 of 8.
Nightshift	The only model provider. Every call logs the model that answered.
NVIDIA Nemotron	nemotron-3.5-lightning-30b-a3b writes the evidence summary. nemotron-3-ultra-550b-a55b is the duty lead.
LangGraph	The desk graph, with the re-query loop and the handover branch.
Cursor	Built the Nightshift provider block (git tag cursor-provider, reviewed in docs/cursor/B-review.md).
What was unique	The experiment's rules live in the sponsor tools: Zetaris cuts every time-stamped table at the issue time, the memory sidecar returns a night only after it was verified, and H-MEM's verified nights are the memory model's training set. Ultra may move the base by at most 0.15, and every number it writes is checked.

SAMPLE_MODE replays recorded model replies and a small data sample, so the desk runs from a fresh clone with no keys.

What the school sees at 18:00



Tonight’s call first, in the plan’s colour; then the probability, as a number and as 100 nights; then the night over Ka‘ū.



A replay of 2 September 2025. Press play and each monitor’s hour rises as a beam; the ring marks 75 ppb.

Live at zeyusuntt.github.io/vog-desk/web. Drag to turn the map, pick a viewpoint, click a monitor for its night.

What it cannot do yet

Limits

- One station and one threshold, 75 ppb in any hour.
- An hourly mean does not prove a 15-minute exposure.
- One training year, with 21 events.
- Warnings on 87 of 102 nights. In a quieter season the probability is the better guide.
- The review loop’s value is not established.
- Ultra fell back to the code’s template on 23 of 102 nights.
- A defect in the recent-rate stream was found after scoring. It did not touch the scored probability and is fixed for the live desk.

Next

- Replicate at Ocean View.
- Score 15-minute exposure from EPA’s 5-minute data.
- Handle feed gaps: the state’s stations appear to post 4 to 6 hours late.
- Add years as the eruption continues.

Zetaris reflection

Where it sped us up

- One query language over Parquet and Postgres, so the evidence code is one module.
- The no-lookahead rule became a publication time on every time-stamped table and one rewrite function.
- A data quality rule flagged exactly the two censored Ocean View hours the state marks.

How we would use it at work

As the contract between forecasters and data engineers. The semantic layer defines what agents may read, and the rules record what the engineers promise about it.

What it should improve

1. A service named postgres shadows the platform’s own database.
2. REGISTER DATASOURCE TABLES scans only the public schema.
3. Reserved column names fail with a generic error.
4. NOT NULL cannot be activated from Parquet.
5. Descriptions reject colons and parentheses.
6. The REST route is undocumented and returns strings.
7. A count takes 4 to 8 s, a DQ run 27 to 35 s.
8. SHOW TABLES is disabled on this release.

Full text: docs/ZETARIS_REFLECTION.md. Local Docker install, 23 October 2026.

vog-desk

From 23 to 26 October 2026 it issued a forecast for Pāhala every evening at 18:00 and checked it against the monitor the next afternoon.

scripts/prove.py re-checks every number shown here from the committed files.

Live page: zeyusuntt.github.io/vog-desk/web

Repository: github.com/zeyusuntt/vog-desk. Video: youtu.be/EGJ8WhEMQvc. Team dotisacat, Open Agent Hackathon 2026, Track 3.